Gazebo Ackermann Vehicle Topics Analysis

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1 Introduction

In this experiment, a simulated Ackermann steering vehicle was launched in Gazebo. The simulator publishes several communication topics that allow external software such as ROS2 to control the vehicle and monitor its state. These topics transmit commands, vehicle motion data, and coordinate transformations.

2 Vehicle Topics

The following table lists the most important topics observed in the simulation.

Topic	Message Type	Purpose
/clock	rosgraph_msgs/Clock	Simulation time
/model/vehicle_blue/cmd_vel	geometry_msgs/Twist	Vehicle motion command
/model/vehicle_blue/odometry	nav_msgs/Odometry	Vehicle position and velocity
/model/vehicle_blue/tf	tf2_msgs/TFMessage	Coordinate transformations

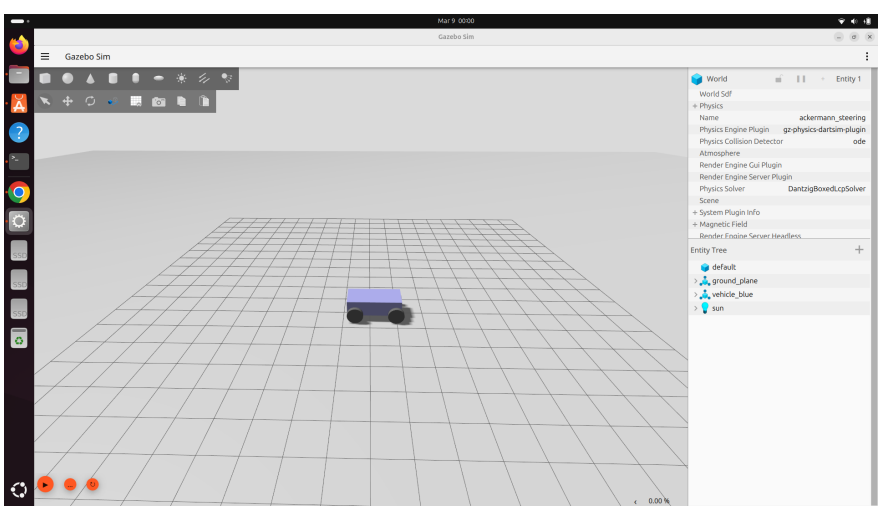

Figure 1: Ackermann steering vehicle in Gazebo simulation

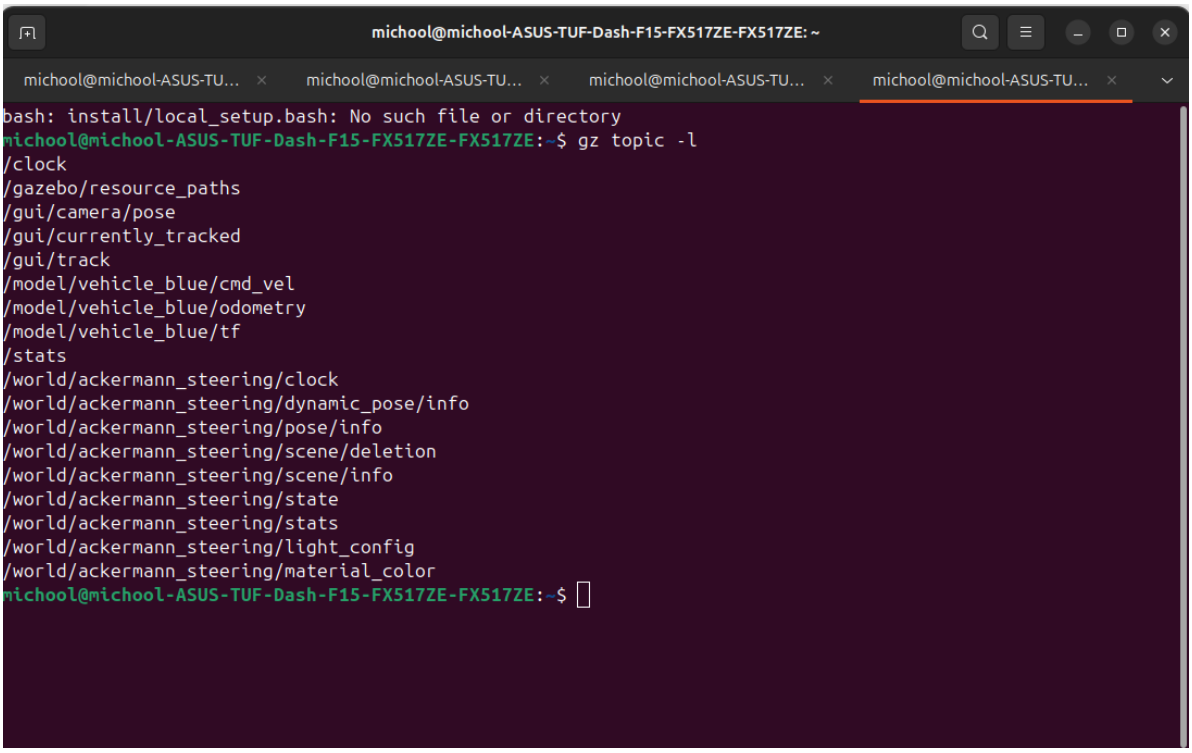

Figure 2: Vehicle Topics in Gazebo Sim

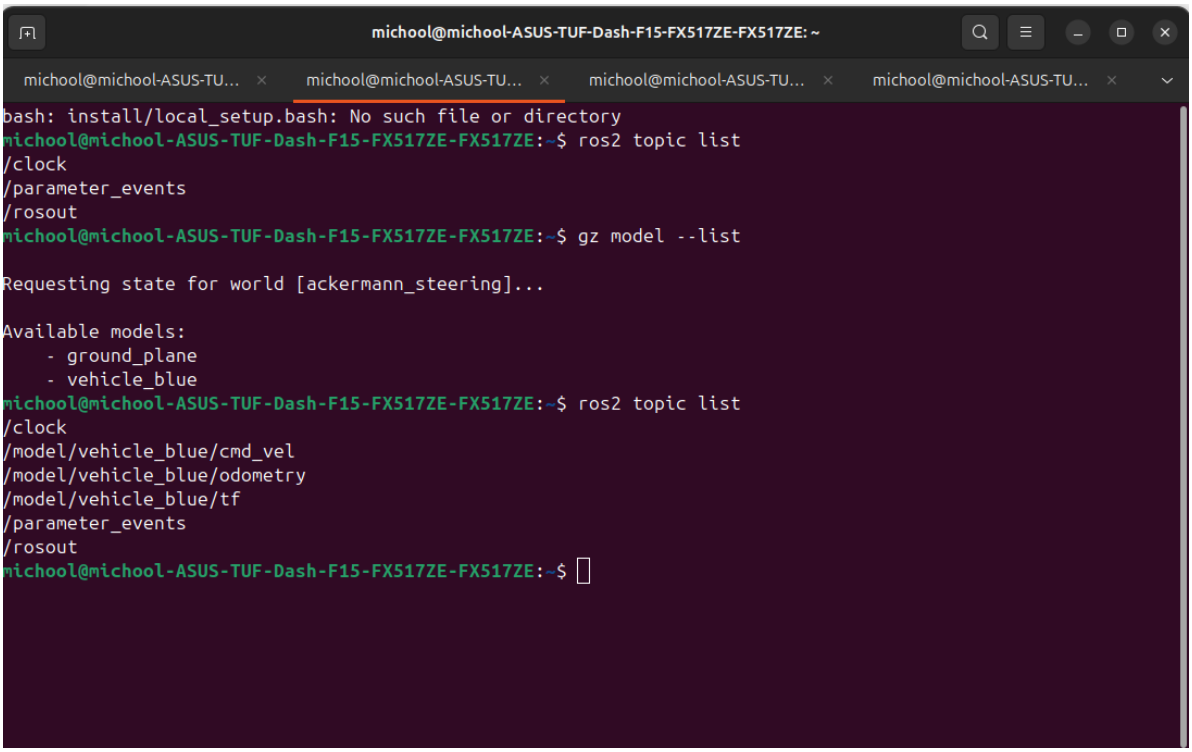

Figure 3: ROS Topics That can be Published and Subscribed